

SANHORN CHEN

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EDUCATION

University of Illinois - Urbana-Champaign

August 2022 - May 2026

Bachelor's, Computer Science + Economics

GPA: 3.66

- Courses: Artificial Intelligence, Computer Vision, Natural Language Processing, Applied Machine Learning, Data Structures, Statistic & Probability, International Business, Economic Forecasting, Financial Economics, Bayesian Machine Learning

PROFESSIONAL EXPERIENCE

Taiwan Semiconductor Manufacturing Company (TSMC)

Hsinchu City, Taiwan

Data Engineer Intern | CPO AI4BI Division

May 2025 - August 2025

- Optimized Python-based ML pipeline for Fab's AI Wafer Output Forecast, leveraging XGBoost with grid search to improve cycle time and output forecasting.
- Transitioned hardcoded scripts into reproducible code aligned with TSMC's MLOps architecture, integrated CI/CD enhancements, and developed unit tests to ensure maintainable and reliable model retraining.
- Increased Fab's AI Wafer Output Forecast's Analysis efficiency by 10x through designing Star Schemas, cleaning and loading datasets through MariaDB API, and writing DAX formulas for PowerBI.

Applied Technologies For Learning in Arts & Sciences, University of Illinois - Urbana Champaign

Urbana, IL, USA

AI/ML Intern

January 2025 - May 2025

- Developed a LLM-based student advising chatbot using GPT-4o-mini leveraging LlamaIndex RAG with Prompt Engineering to enhance recommendation capabilities tailored to students' interests.
- Engineered full-stack deployment pipeline for the advising chatbot by developing frontend using HTML, CSS, and JavaScript, building a FastAPI backend, and integrating the LlamaIndex vector store with GPT-4o-mini

Applied Technologies For Learning in Arts & Sciences, University of Illinois - Urbana Champaign

Urbana, IL, USA

Data Analyst Intern

August 2024 - December 2024

- Developed interactive Tableau dashboards to visualize UIUC computer classroom datasets, uncovering key usage trends and supporting data-driven decision-making across resource allocation and planning.
- Enhanced analytical efficiency by integrating API-based data extraction and performing comprehensive data cleaning, restructuring, and feature engineering on unstructured datasets using Python, NumPy, and Pandas.

Foxconn (Hon Hai) Research Institute

Taipei City, Taiwan

AI Research Intern | Computer Vision Group

June 2023 - February 2024

- Proposed a model (CARGAN) that outputs street view images for driving datasets, preserving the street content while varying the weather style to generate images in different weather conditions such as sunny, snowy, rainy, and overcast using Pytorch.
- Developed 3 novel loss functions (Source Preserving Loss, Style Similarity Loss, and Content Contrastive Loss) to enhance the I2I translation results and effectively preserve the content of the original source.
- Organized 7 different weather conditions' street view dataset from BDD100K with Python and Javascript. Structured the dataset into training and testing sets in a format compatible with StarGAN V2.

RESEARCH EXPERIENCE

iDEA-iSAIL Joint Laboratory, University of Illinois - Urbana Champaign

Urbana, IL, USA

Undergraduate Research Assistant Guided By Prof. HangHang Tong

March 2025 - Present

- Conducted Time-Series Question Answering (QA) research and co-first authored a paper, Time Series Analysis Question Answering Benchmark, preparing for submission to ACL 2026, introducing a novel benchmark that evaluates reasoning abilities of time series foundation models, LLMs, and VLMs on proposed benchmark.
- First authored a survey paper on Time-Series Question Answering preparing for submission to IJCAI 2026, conducted an in depth study of QA methodologies, time series foundation models for natural language tasks, and benchmark datasets, and proposing limitations and future research directions.
- Engineered complete pipelines for data preprocessing, PyTorch zero shot evaluation, model quantization, and LoRA instruction fine tuning, and executed extensive experiments on A100 GPUs.

Social Computing Systems Lab, University of Illinois - Urbana Champaign

Urbana, IL, USA

Undergraduate Research Assistant Guided By Prof. Yun Huang

March 2025 - Present

- Designed and developed a multi-agent IEP meeting simulation system using Python and LangChain's multi-agent framework, integrating GPT api to enable realistic role-based interactions among agents representing patients, family members, and therapists.
- Contributed to Speech-Language Pathologist (SLP) training and education by providing therapists with an interactive multi-agent platform for pre-session preparation, new-staff training, and real-world meeting practice.
- Mentored students in IS492: Designing Human-Centered AI Systems, providing guidance and recommendations on developing an AI Tutor to enhance personalized learning and feedback.

The Adaptive Cognition and Interaction Design Lab, University of Illinois - Urbana Champaign

Urbana, IL, USA

Undergraduate Research Assistant Guided By Prof. Jessie Chin

August 2024 - Present

- Enhancing a React.js web-app using Socket.IO and REST API, focusing on improving user interaction and experience in human-computer interaction.
- Deploying the online board game web-app via Firebase Hosting for client side and Render with PostgreSQL for server side.
- Integrating Large Language Models to replace computer players, enhancing human-computer interaction by enabling more dynamic and adaptive gameplay experiences.

Econ Data Lab, University of Illinois - Urbana Champaign

Urbana, IL, USA

Undergraduate Research Assistant Guided By Prof. Daniela Alonso Fontes

January 2025 - May 2025

- Cleaned and restructured large-scale panel time series data from IFLS4 and IFLS5 using Python pandas and NumPy to analyze provincial unemployment and sector-specific employment trends across Indonesia.
- Investigated urban unemployment dynamics during expansionary business cycles, using data visualizations to examine the impact of internal migration on labor market outcomes in urban provinces.

Department of Computer Science, City University of HongKong

Kowloon, Hong Kong S.A.R.

Summer Research Intern Guided By Prof. Li Min Ming

June 2024 - August 2024

- Led a project on Public Event Scheduling with Busy Agents, incorporating the "Longest Viable Gap" algorithm under Prof. Li Min Ming's guidance to minimize total disagreement through innovative solutions.
- Developed and implemented approximation algorithms with high approximation ratios to effectively reduce total disagreement.

PUBLICATIONS

- **Sanhorn Chen***, Baoyu Jing*, Lecheng Zheng, Boyu Liu, Zihao Li, Jiaru Zou, Tianxin Wei, Zhining Liu, Zhichen Zeng, Ruizhong Qiu, Xiao Lin, Yuchen Yan, Qi Yu, Dongqi Fu, Jingchao Ni, Jingrui He, and Hanghang Tong. "TSAQA: Time Series Analysis Question and Answering Benchmark." Submitted to The Annual Meeting of the Association for Computational Linguistics (ACL), 2026.
- **Sanhorn Chen**, Baoyu Jing, Zeya Wang, Hanghang Tong. "A Survey on Time Series Question Answering." To be submitted to International Joint Conference on Artificial Intelligence (IJCAI), 2026.
- Ziyang Lin, Zixuan Sun, **Sanhorn Chen**, Xiaoyang Chen, Roy Zhao "Accelerating Multi-modal LLM Gaming Performance via Input Prediction and Mishit Correction." *arXiv preprint*, 2025.

PATENTS

- Shen-Hsuan Liu, Chi-En Huang, Muhammad Saqlain Aslam, Yung-Hui Li, **Sanhorn Chen** "Method and Apparatus Related to Data Generation Framework." *US Patent Application* 19/069,248, 2025.
- Shen-Hsuan Liu, Chi-En Huang, Muhammad Saqlain Aslam, Yung-Hui Li, **Sanhorn Chen** "Method and Apparatus Related to Data Generation Framework." *US Patent Application* 19/069,242, 2025.

LEADERSHIPS

Taiwanese Student Association, University of Illinois - Urbana Champaign

Urbana, IL, USA

President

August 2024 - August 2025

- Led weekly meetings with 7 department directors and monthly meetings with 30+ members.
- Collaborated with companies such as TSMC, State Farm, and local restaurants to host inter- and intra-school events, including new student orientation, career talks, company information sessions.

Gies Capital Association, University of Illinois - Urbana Champaign

Urbana, IL, USA

Vice President of Technology

August 2024 - Present

- Designed and hosted technical workshops covering data preprocessing, exploratory data analysis, and visualization pipelines.

- Guided student teams through quantitative research studies, emphasizing reproducibility, model interpretation, and applied insights in finance.

SKILLS

Skills: Python, Pytorch, Pandas, NumPy, C/C++, R, SQL, PowerBI, Tableau, Excel

Languages: Mandarin, English